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# PRACTICAL : Basic Statistics - Measures of Central Tendencies and Variance
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# -----
# Cell 1 : Import Libraries
# -----

import pandas as pd
import numpy as np

print("Libraries Imported Successfully")

# -----
# Cell 2 : Load Dataset
# -----

df = pd.read_csv("iris.csv")

print("Dataset Loaded Successfully")
print(df.head())

# -----
# Cell 3 : Dataset Information
# -----

print("\nDataset Information")
print(df.info())

print("\nBasic Statistical Summary")
print(df.describe())

# -----
# Cell 4 : Check Column Names
# -----

print("\nColumn Names")
print(df.columns)

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# PART 1 : Summary Statistics Grouped by Categorical Variable
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# Cell 5 : Grouped Summary Statistics
# -----

group_stats = df.groupby('species')[[
    'sepal_length',
    'sepal_width',
    'petal_length',
    'petal_width'
]].agg(['mean', 'median', 'min', 'max', 'std'])

print("\nGrouped Summary Statistics")
print(group_stats)

# -----
# Cell 6 : Create List of Numeric Values for Each Category
# -----

group_list = df.groupby('species')['sepal_length'].apply(list)

print("\nList of Sepal Length Values for Each Species")
print(group_list)

# =====
# PART 2 : Statistical Details of Each Species
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# -----
# Cell 7 : Filter Dataset According to Species
# -----

setosa = df[df['species'] == 'setosa']
versicolor = df[df['species'] == 'versicolor']
virginica = df[df['species'] == 'virginica']

print("\nDatasets Filtered Successfully")

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# Cell 8 : Basic Statistical Details
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```
print("\n===== SETOSA =====")
print(setosa.describe())
```

```
print("\n===== VERSICOLOR =====")
print(versicolor.describe())
```

```
print("\n===== VIRGINICA =====")
print(virginica.describe())
```

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# Cell 9 : Percentile Calculation
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```
num_cols = df.select_dtypes(include='number').columns
```

```
print("\nSetosa Percentiles")
print(setosa[num_cols].quantile([0.25, 0.50, 0.75]))
```

```
print("\nVersicolor Percentiles")
print(versicolor[num_cols].quantile([0.25, 0.50, 0.75]))
```

```
print("\nVirginica Percentiles")
print(virginica[num_cols].quantile([0.25, 0.50, 0.75]))
```

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# Cell 10 : Standard Deviation
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```
print("\nSetosa Standard Deviation")
print(setosa[num_cols].std())
```

```
print("\nVersicolor Standard Deviation")
print(versicolor[num_cols].std())
```

```
print("\nVirginica Standard Deviation")
print(virginica[num_cols].std())
```

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# Cell 11 : Mean Values
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```
print("\nSetosa Mean")  
print(setosa[num_cols].mean())
```

```
print("\nVersicolor Mean")  
print(versicolor[num_cols].mean())
```

```
print("\nVirginica Mean")  
print(virginica[num_cols].mean())
```

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# Cell 12 : Median Values
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```
print("\nSetosa Median")  
print(setosa[num_cols].median())
```

```
print("\nVersicolor Median")  
print(versicolor[num_cols].median())
```

```
print("\nVirginica Median")  
print(virginica[num_cols].median())
```

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# END OF PRACTICAL
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